



102 MILES

THE LENGTH OF THE SUEZ CANAL WHEN IT WAS FIRST BUILT

(1824–1907) both noticed a bright yellow line at a wavelength that did not match any known substance. Lockyer and Janssen suggested that this indicated an unknown element in the Sun, which Lockyer called **helium**, after the Greek “helios” for Sun. That year, Swedish physicist Anders Ångström (1814–74) also mapped the complete spectrum



DMITRI MENDELEEV
(1834–07)

Born in 1834 in Tobolsk, Siberia, Dmitri Mendeleev hiked to St. Petersburg to enroll in the university. Despite suffering from tuberculosis, he became Russia's leading chemist and developed the Periodic Table, which earned him worldwide acclaim and enabled him to predict the discovery of the elements gallium, scandium, and germanium.

of light from the Sun, identifying a thousand spectral lines in units that became known as angstroms in his honor.

The debate about evolution continued. Darwin described how traits might be passed on from generation to generation through a process called **pangeneses**. He proposed that in the body there are countless particles, or gemmules, which are like seeds that can reproduce the whole organism. Some claim this has similarities with DNA, the genetic material found in every body cell, which was coincidentally identified in cell nuclei the following year by Swiss biology student Friedrich Miescher (1844–95). It is now known that only DNA in the sex, or germ, cells (eggs and sperm) is ever used to make a new organism.

In France, geologist **Louis Lartet** (1840–99) added weight to the idea that humans also evolved, with his discovery of the first identified skeletons of what came to be known as **Cro-Magnon man**. They were named after the cave near Les

Eyzies in France where the remains were found. They are now more generally known as European Early Modern Humans.

More controversially, in 1869 Darwin's half-cousin **Francis Galton** (1822–1911) used Darwin's theories to suggest a hereditary basis for human intelligence, an idea that was to lead him to develop the **science of eugenics**.

Technological change gathered pace as British inventor **John Peake Knight** (1828–86) invented **traffic lights**, and in the US American engineer **George Westinghouse** (1846–1914) invented **air brakes**. Another American, inventor **John Hyatt** (1837–1920), developed **celluloid**.

The **Suez Canal opened** to shipping in November 1869, linking the Red Sea to the

Table of elements
This Russian periodic table is based on Mendeleev's original table. It includes his predicted elements—gallium, scandium, and germanium.

Mediterranean. When it first opened the canal was 102 miles (164 km) long. It took 10 years of construction work to complete.

PERИОДИЧЕСКАЯ СИСТЕМА ЭЛЕМЕНТОВ Д.И.МЕНДЕЛЕЕВА																		VII		VIII	
1	H				II		III		IV		V		VI				1	2	He		
	1.00794				4.002602												1.00794	4.002602	Гелий		
2	Li		Be		3		4		5		6		7		8		9	10	Ne		
	6.941		9.01218		БЕРИЛЛИЙ				10.81		12.011		14.0067		15.999		16.99944	20.1797	Неон		
3	Na		Mg		11		12		13		14		15		16		17	18	Ar		
	22.98977		24.305		МАГНИЙ		25.98256		28.086		30.97376		32.06		35.453		39.948	39.948	Аргон		
4	K		Ca		19		20		21		22		23		24		25	26	Co		
	39.098		40.078		КАЛЬЦИЙ		44.9559		47.90		50.9418		51.996		54.93809		55.845	58.9332	Никель		
5	Rb		Sr		37		38		39		40		41		42		43	44	Pd		
	85.4678		87.62		СТРОНЦИЙ		88.9058		91.224		92.90638		95.94		98.9062		101.07	106.42	Палладий		
6	Ag		Cd		47		48		49		50		51		52		53	54	Xe		
	107.8682		112.411		КАДМИЙ		114.904		118.710		121.757		127.6		126.90545		131.29	131.29	Ксенон		
7	Cs		Ba		55		56		57		58		59		60		61	62	Rn		
	132.90545		137.327		БАРИЙ		138.90547		174.967		176.43		183.84		186.207		222	222	Радон		
8	Au		Hg		79		80		81		82		83		84		85	86	U		
	196.96657		200.5958		РУТУИЙ		204.38		207.2		208.9804		209		210		238.02891	238.02891	Уран		
9	Fr		Ra		87		88		89		90		91		92		93				
	(223)		226.0254		РАДИЙ		227		232		238.02891		244		244		244				
* ЛАНТАНОИДЫ *																					
10	Ce		Pr		58		59		60		61		62		63		64		65		
	140.12		140.9077		НЕОДИМ		144.24		(145)		150.4		151.96		157.25		158.9254		162.5		
11	Gd		Tb		64		65		66		67		68		69		70		71		
	157.25		158.9254		ДИПРОЗИЙ		158.9254		162.5		162.5		164.9304		167.25		168.9304		171.937		
12	Ho		Er		69		70		71		72		73		74		75		76		
	164.9304		167.25		ЭРБИЙ		168.9304		171.937		173.0547		174.967		176.9304		178.9304		180.9304		
13	Tm		Yb		72		73		74		75		76		77		78		79		
	168.9304		171.937		ТУЛИЙ		173.0547		174.967		176.9304		178.9304		180.9304		182.9304		184.9304		
14	Lu				71																
	174.967				ЛУТЦИЙ																
* АКТИНОИДЫ *																					
15	Th		Pa		90		91		92		93		94		95		96		97		
	232.0377		231.03688		ПРОАКТИНИЙ		238.02891		237.04817		244		244		244		244		244		
16	U		Np		92		93		94		95		96		97		98		99		
	238.02891		237.04817		УРАН		244		244		244		244		244		244		244		
17	Pu		Am		94		95		96		97		98		99		100		101		
	244		244		АМЕРИЦИЙ		244		244		244		244		244		244		244		
18	Cm		Bk		96		97		98		99		100		101		102		103		
	244		244		КУРИЙ		244		244		244		244		244		244		244		
19	Cf		Es		98		99		100		101		102		103		104		105		
	244		244		КАЛИФОРНИЙ		244		244		244		244		244		244		244		
20	Fm		Md		100		101		102		103		104		105		106		107		
	244		244		ФЕРМИЙ		244		244		244		244		244		244		244		
21	No		Lr		102		103		104		105		106		107		108		109		
	244		244		НОБЕЛИЙ		244		244		244		244		244		244		244		